



Unit 3 — Mathematics for Artificial Intelligence

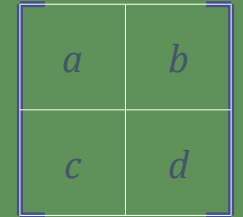
XI • UNIT 3 • CHAPTER 1

Matrices

Introduction to Matrices • Types of Matrices

Matrix Operations: Addition, Subtraction, Multiplication, Transpose

Robotics and AI Dept



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What We Will Cover



01

Introduction to Matrices

Definition, order ($m \times n$), notation and elements



02

Types of Matrices

Row, column, square, diagonal, scalar, identity, zero



03

Addition & Subtraction

Element-by-element operations on matrices of the same order



04

Multiplication

Matrix-matrix rule, scalar multiplication and key properties



05

Transpose of a Matrix

Definition, properties, symmetric & skew-symmetric matrices

Introduction to Matrices

A matrix is a rectangular array of numbers or functions arranged in rows and columns, enclosed in square brackets [] or parentheses (). The individual numbers are called its elements or entries.

Order of a Matrix

A matrix with m rows and n columns is an $m \times n$ matrix (“ m by n ”) — this is called its order.

Row matrix $\rightarrow 1 \times n$ *Column matrix* $\rightarrow m \times 1$

Example

A

1	3
2	4
5	6

A is a matrix of order 3×2 .

Row 1: 1, 3 Row 2: 2, 4 Row 3: 5, 6

Column 1: 1, 2, 5 Column 2: 3, 4, 6

General Form & Notation

General $m \times n$ Matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The element in the i th row and j th column of matrix A is written a_{ij} . Rows are counted top to bottom; columns left to right.



Reading the Notation

- a_{12} means the element in row 1, column 2.
- The first subscript is always the row number.
- The second subscript is always the column number.
- A matrix of order $m \times n$ has exactly $m \times n$ elements.

Types of Matrices (I)

Row Matrix

Only one row.

$$\begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$$

Column Matrix

Only one column.

$$\begin{bmatrix} 3 \\ 7 \\ 9 \end{bmatrix}$$

Square Matrix

Rows equal columns.

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Diagonal Matrix

Nonzero only on the diagonal.

$$\begin{bmatrix} 5 & 0 \\ 0 & 8 \end{bmatrix}$$

Types of Matrices (II)

Scalar Matrix

A diagonal matrix in which all diagonal elements are equal.

$$\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

Identity Matrix (I)

Square matrix with 1's on the diagonal and 0's elsewhere.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Zero (Null) Matrix

Every element of the matrix is zero.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Addition & Subtraction

Rule: matrices must be of the same order. Add or subtract corresponding elements.



Addition (A + B)

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix}$$

Add corresponding elements:

$$1+5=6, 3+7=10, 2+6=8, 4+8=12$$

Result is the same order (2×2) as A and B.



Subtraction (A - B)

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

Subtract corresponding elements:

$$1-5=-4, 3-7=-4, 2-6=-4, 4-8=-4$$

Subtraction is only defined for matrices of the same order.

Matrix Multiplication



Compatibility Rule

An $m \times n$ matrix can be multiplied by an $n \times k$ matrix only when the number of columns in the first equals the number of rows in the second. The result is an $m \times k$ matrix.

$$(m \times n) \times (n \times k) = (m \times k)$$

Element (i, j) of the product = sum of (row i of $A \times$ column j of B).

Worked Example

$$A \times B = AB$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 8 & 8 \end{bmatrix}$$

$$\begin{aligned} (1,1): 1 \cdot 2 + 3 \cdot 1 &= 5 & (1,2): 1 \cdot 0 + 3 \cdot 2 &= 6 \\ (2,1): 2 \cdot 2 + 4 \cdot 1 &= 8 & (2,2): 2 \cdot 0 + 4 \cdot 2 &= 8 \end{aligned}$$

Key Properties

- Not commutative in general: $AB \neq BA$
- Associative: $(AB)C = A(BC)$

Scalar Multiplication

A matrix can be multiplied by a scalar (a single number). The rule is simple: multiply every element of the matrix by that scalar.

General Rule

$$k \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} k & 3k \\ 2k & 4k \end{bmatrix}$$

Example: $3 \times [2 \ -1; 4 \ 0] = [6 \ -3; 12 \ 0]$ — every entry is scaled by 3.



Key Properties Recap

- Matrix multiplication is NOT commutative: $AB \neq BA$ (in general)
- Matrix multiplication IS associative: $(AB)C = A(BC)$
- Scalar multiplication distributes over every element

Transpose of a Matrix

The transpose of a matrix A , written A^T , is obtained by interchanging its rows and columns: rows become columns and columns become rows. In notation, $(A^T)_{ij} = A_{ji}$.

If A is $m \times n$, then A^T is $n \times m$ — the order is reversed.



Think of it as flipping the matrix along its main diagonal.

Worked Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$A \ (2 \times 3)$



$$\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$A^T \ (3 \times 2)$

Row 1 of $A \rightarrow$ Column 1 of A^T . Row 2 of $A \rightarrow$ Column 2 of A^T .

Transpose Properties & Special Matrices

Key Properties

- $(A^T)^T = A$ — transposing twice returns the original matrix
- $(A + B)^T = A^T + B^T$ — transpose of a sum is the sum of transposes
- $(AB)^T = B^T A^T$ — transpose of a product reverses the order

Symmetric

A square matrix where $A^T = A$. Elements mirror across the main diagonal ($a_{ij} = a_{ji}$).

$$\begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$$

Skew-symmetric

A square matrix where $A^T = -A$. Diagonal elements are 0, and $a_{ij} = -a_{ji}$.

$$\begin{bmatrix} 0 & b & c \\ -b & 0 & d \\ -c & -d & 0 \end{bmatrix}$$

Thank You

Questions & Discussion

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